

MODULE 2

• Machine learning - family of statistical modelling techniques

• Systems - improve - with ↑ - improve accuracy and speed techniques / approaches

=> Classified in a spectrum of supervision
automatically learn and improve PREDICTIONS (without programming)

Supervised learning:

process through which a program takes input-output pairs and "learns" the function that links them together

=> function → models

defining algorithms - models (functions) that matches well inputs - outputs

Training set
examples

How is the function right?

Objective → model works on unseen examples

Validation set

probes / test

periodically

we learn from here and tweak hyper parameters

hyperparameters
external configuration parameters that a data scientist sets before training a model.

They control various aspects of the learning process and model's architecture, influencing how well the model generalizes to new data.

The model doesn't learn from validation set but we use that to tweak the parameters.

Test set
try out

TRAINING LOOP

hyperparameters tuning



until performance is not better

How can unstructured data represent a model?

images → pixels are numerical

unstructured data → words → vectors



word embedding

Regression

Classification

Real numbers

Categories

Model training =

minimizing difference between output of the model's function and the true label for every sample in the training set

function - True label = Loss

output

numerical
value

↓
We average the loss

Then the model updates its function to map accurately

fitting / training the model

Adapts parameters → numerical values that are used to perform operations on the input data

Bias ← concept around model performance and algorithm fairness
descriptor for additive parameters

"weights"

learnable parameters that determine the strength of connections between neurons or features in a model

LOGISTIC REGRESSION

used in → Categories → binaries



Sigmoid function

Non-linear transformation → all values between 0-1

probabilities

Probability cut off = threshold
decision boundary or operating point

Approaches to supervised learning

- logistic regression
- linear " / polynomial
- decision trees
- support vector machines
- neural networks

"No free lunch" theorem

Regression Tweaks $\rightarrow$ "regularization methods" - with approaches to
better accommodate dataset challenge

Decision Trees

Random forest - $\neq$ criteria

Supervised Vector Machine (for classification)

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Work better w/ crazy outliers
multidimensional features

Unsupervised learning
finds patterns / not labels



in a dataset based on the structure of the data (not on pre-determined labels)

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